

Lab Work 2
Course Title: Information Security Laboratory/Project
Course Code: 0714 02 CSE 4108
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1. Implementation of RSA

Source code:

```
def gcd(a, b):  
    while b != 0:  
        temp = b  
        b = a % b  
        a = temp  
    return a  
  
def mod_exp(base, exp, mod):  
    result = 1  
    base = base % mod  
    while exp > 0:  
        if exp % 2 == 1:  
            result = (result * base) % mod  
        exp = exp >> 1  
        base = (base * base) % mod  
    return result  
  
def mod_inverse(e, phi):  
    for d in range(1, phi):  
        if (e * d) % phi == 1:  
            return d
```

```
return -1
```

```
def main():
```

```
    print("==== RSA Cryptosystem Implementation =====")
```

```
    p = int(input("Enter prime number p: "))
```

```
    q = int(input("Enter prime number q: "))
```

```
    n = p * q
```

```
    phi = (p - 1) * (q - 1)
```

```
    print("Computed n =", n)
```

```
    print("Computed phi (n) =", phi)
```

```
    e = int(input("\nEnter public exponent e (gcd(e, phi) = 1): "))
```

```
    if gcd(e, phi) != 1:
```

```
        print("Invalid e! gcd(e, phi) must be 1.")
```

```
        return
```

```
    d = mod_inverse(e, phi)
```

```
    if d == -1:
```

```
        print("Modular inverse not found!")
```

```
        return
```

```
    print("Public Key (e, n):", (e, n))
```

```
    print("Private Key (d, n):", (d, n))
```

```
    while True:
```

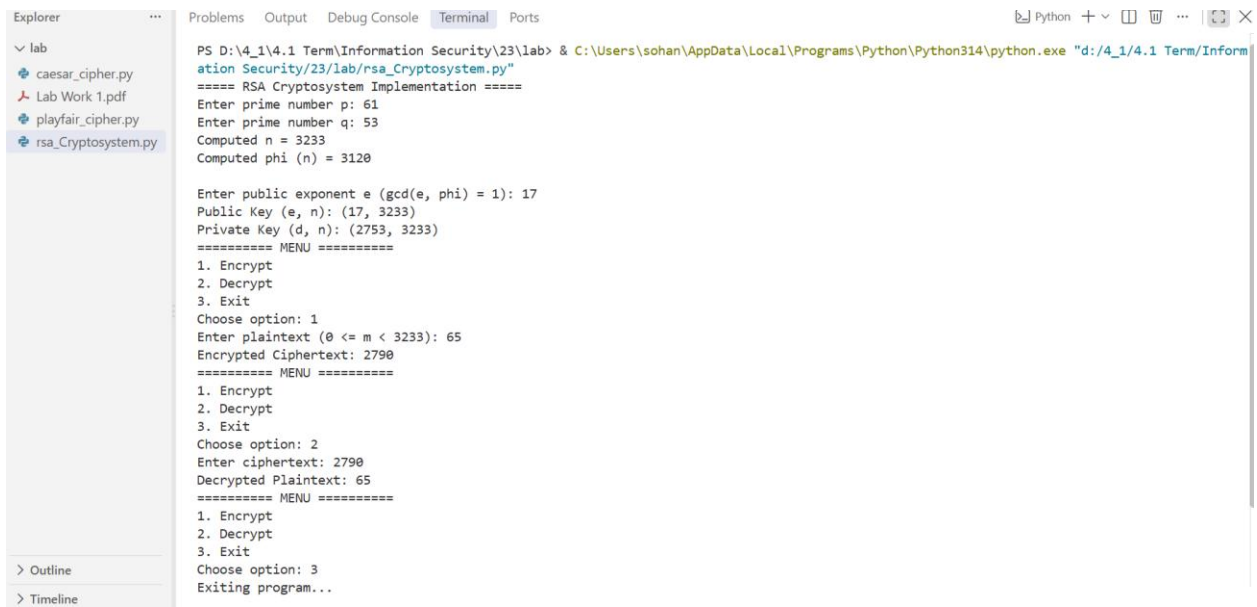
```
        print("===== MENU =====")
```

```
        print("1. Encrypt")
```

```
        print("2. Decrypt")
```

```
print("3. Exit")
choice = int(input("Choose option: "))
if choice == 1:
    m = int(input(f"Enter plaintext ( $0 \leq m < \{n\}$ ): "))
    if m < 0 or m >= n:
        print("Invalid message range!")
        continue
    C = mod_exp(m, e, n)
    print("Encrypted Ciphertext:", C)
elif choice == 2:
    C = int(input("Enter ciphertext: "))
    m = mod_exp(C, d, n)
    print("Decrypted Plaintext:", m)
elif choice == 3:
    print("Exiting program...")
    break
else:
    print("Invalid choice! Try again.")
if __name__ == "__main__":
    main()
```

Output:



```
PS D:\4_1\4.1 Term\Information Security\23\lab> & C:\Users\sohan\AppData\Local\Programs\Python\Python314\python.exe "d:/4_1/4.1 Term/Information Security/23/lab/rsa_Cryptosystem.py"
===== RSA Cryptosystem Implementation =====
Enter prime number p: 61
Enter prime number q: 53
Computed n = 3233
Computed phi (n) = 3120

Enter public exponent e (gcd(e, phi) = 1): 17
Public Key (e, n): (17, 3233)
Private Key (d, n): (2753, 3233)
===== MENU =====
1. Encrypt
2. Decrypt
3. Exit
Choose option: 1
Enter plaintext (0 <= m < 3233): 65
Encrypted Ciphertext: 2790
===== MENU =====
1. Encrypt
2. Decrypt
3. Exit
Choose option: 2
Enter ciphertext: 2790
Decrypted Plaintext: 65
===== MENU =====
1. Encrypt
2. Decrypt
3. Exit
Choose option: 3
Exiting program...
```